The full Burnside–Stirling–Gautschi bound

MathDB 372858: solution of Schmidlin's Conjecture 37

Principia Math In Progress

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Task 1. Prove the bound on its full stated domain

For $x \geq 0$ and $0 \leq \alpha \leq 1$, put

$$\widehat{\Pi}(x,\alpha) = \Gamma(x+1)(x+1)^{\alpha}, \qquad S_{1/2}(y) = \sqrt{2\pi}(y+\tfrac{1}{2})^{y+1/2}e^{-y-1/2},$$

and $\widehat{m}_{1/2}(x,\alpha) = \log\big(\widehat{\Pi}(x,\alpha)/S_{1/2}(x+\alpha)\big)$. These are the definitions of [1], pp. 3–4 and 10. The domain $\widehat{D}_{1/2}$ is exactly $[0,\infty)\times[0,1]$. Conjecture 37, pp. 28–29, asks to remove the restriction $x \geq 1$ in Corollary 32.

Theorem 1. *For every $x \geq 0$ and $\alpha \in [0,1]$,*

$$\widehat{m}_{1/2}(x,\alpha) < \frac{1}{12(x+\alpha)+18-12e^{1/48}}. \tag{1}$$

In particular, Conjecture 37 holds, with strict inequality.

We first record the only special-function estimate needed.

Lemma 1. *For every $u > 0$, the Stirling remainder*

$$r(u) = \log\Gamma(u) - (u-\tfrac{1}{2})\log u + u - \tfrac{1}{2}\log(2\pi)$$

satisfies

$$r(u) < B(u) := \frac{1}{12u} - \frac{1}{360u^3} + \frac{1}{1260u^5}. \tag{2}$$

Proof. Binet's formula [2, Eq. 5.9.10] gives

$$r(u) = 2\int_0^{\infty} \frac{\arctan(t/u)}{e^{2\pi t}-1}\,dt.$$

The exact identity

$$z - \frac{z^3}{3} + \frac{z^5}{5} - \arctan z = \int_0^z \frac{s^6}{1+s^2}\,ds > 0 \qquad (z>0)$$

holds on the whole positive axis. Also,

$$\int_0^{\infty} \frac{t^j}{e^{2\pi t}-1}\,dt = \frac{j!\,\zeta(j+1)}{(2\pi)^{j+1}} = \begin{cases} 1/24 & j=1,\\ 1/240 & j=3,\\ 1/504 & j=5. \end{cases}$$

The first equality follows by expanding the positive geometric series and integrating termwise. Integrating the strict arctangent bound proves (2); every integral involved is finite. □

Proof of the theorem. Set $u = x + 1 \geq 1$, $\delta = \alpha - \frac{1}{2} \in [-\frac{1}{2}, \frac{1}{2}]$, and $d = 12(e^{1/48} - 1)$. The elementary inequalities $1 + t < e^t < (1 - t)^{-1}$, for $0 < t < 1$, imply

$$\frac{1}{4} < d < \frac{12}{47} < \frac{1}{3}. \tag{3}$$

Direct substitution in the definitions gives

$$\widehat{m}_{1/2}(x, \alpha) = r(u) + \delta - (u + \delta) \log(1 + \delta/u).$$

Thus we must show that

$$F_u(\delta) := r(u) + \delta - (u + \delta) \log(1 + \delta/u) - \frac{1}{12(u + \delta) - d} < 0.$$

All denominators are positive, since $u + \delta \geq \frac{1}{2}$. Differentiation yields

$$F_u'(\delta) = -\log(1 + \delta/u) + \frac{12}{[12(u + \delta) - d]^2}, \tag{4}$$

$$F_u''(\delta) = -\frac{1}{u + \delta} - \frac{288}{[12(u + \delta) - d]^3} < 0. \tag{5}$$

In particular $F_u'(\delta) > 0$ when $\delta \leq 0$. At $\delta = 1/12$, use $\log(1 + t) > t/(1 + t)$ for $t > 0$. For $u = 1 + y$, $y \geq 0$, one has

$$(12u + \tfrac{2}{3})^2 - 12(12u + 1) = \frac{40}{9} + 160y + 144y^2 > 0.$$

Together with $d < 1/3$, this shows

$$\frac{12}{(12u + 1 - d)^2} < \frac{12}{(12u + 2/3)^2} < \frac{1}{12u + 1} < \log(1 + 1/(12u)).$$

Hence $F_u'(1/12) < 0$. Strict concavity now places the global maximum of F_u on $[-\frac{1}{2}, \frac{1}{2}]$ inside $(0, 1/12)$.

On $0 \leq \delta \leq 1/12$, let

$$v = u + \tfrac{1}{12}, \qquad K(u) = \frac{1}{v} + \frac{1}{6v^3} = \frac{12(144u^2 + 24u + 25)}{(12u + 1)^3} > 0.$$

Because $d > 0$ and $u + \delta \leq v$, (5) implies $F_u''(\delta) \leq -K(u)$. Integrating twice and completing a square,

$$F_u(\delta) \leq F_u(0) + F_u'(0)\delta - \tfrac{1}{2}K(u)\delta^2 \leq F_u(0) + \frac{F_u'(0)^2}{2K(u)}.$$

Lemma 1 and (3) give

$$F_u(0) < B(u) - \frac{1}{12u - 1/4}, \qquad 0 < F_u'(0) < \frac{12}{(12u - 1/3)^2}.$$

Consequently the global maximum of F_u is strictly less than $-R(u)$, where

$$R(u) = \frac{1}{12u - 1/4} - \frac{1}{12u} + \frac{1}{360u^3} - \frac{1}{1260u^5} - \frac{72}{(12u - 1/3)^4 K(u)}. \tag{6}$$

It remains to prove one rational inequality, $R(u) > 0$ for $u \geq 1$. Write $Q(u) = 144u^2 + 24u + 25$. Exact simplification gives

$$R(u) = \frac{N(u)}{2520u^5(36u-1)^4(48u-1)Q(u)},$$

where the numerator can be formed without any division:

$$\begin{aligned} N(u) = {} & 210u^4(36u-1)^4Q(u) + 7u^2(36u-1)^4(48u-1)Q(u) \\ & - 2(36u-1)^4(48u-1)Q(u) - 1224720u^5(48u-1)(12u+1)^3. \end{aligned}$$

For an explicit positivity certificate, write $N(1+y) = \sum_{j=0}^{10} c_j y^j$. Its integer coefficients are

j	c_j	j	c_j
0	2417815645	6	8758379206944
1	130284291970	7	5353548256512
2	968629141445	8	2115822352896
3	3398691532320	9	490420998144
4	7091567419890	10	50791587840
5	9614662445664		

Every coefficient is strictly positive, so $N(u) > 0$ for $u \geq 1$. The denominator is positive there as well. This proves $R(u) > 0$, then $F_u(\delta) < 0$ on the full required interval, and finally (1). □

Certificate and scope. The accompanying `verify_polynomial.py` builds the numerator from its four displayed summands using exact integers, checks the translation $u = 1 + y$ by both binomial expansion and Horner substitution, and verifies the rational identity at exact rational points. The displayed polynomial identity and positive coefficients certify the infinite range; numerical sampling is not used as a proof. The argument includes $x = 0$, both endpoints of $[0, 1]$, and arbitrarily large x.

Tasks 2–5. Further directions after the full solve

These are proposed tasks, not additional solved claims; each should receive a fresh literature and public-claim check before being attacked.

Task 2. Determine the largest constant C for which $\widehat{m}_{1/2}(x, \alpha) \leq [12(x+\alpha) + C]^{-1}$ holds on the full domain, with a positive denominator, and identify all extremizers or limiting extremizers.

Task 3. For the constant in (1), analyze the unique maximizing δ of F_u: determine its monotonicity in u and derive an asymptotic expansion with an explicit remainder.

Task 4. Extend the same inequality, wherever its denominator is positive, to as much of the natural Gamma-function domain $x > -1,\ 0 \leq \alpha \leq 1,\ x + \alpha > -\frac{1}{2}$ as possible.

Task 5. Find a simpler positive expression below $R(u)$ for all $u \geq 1$, preserving its leading asymptotic term, to replace the degree-ten coefficient certificate by a shorter quantitative bound.

References

[1] M. Schmidlin, *On Gautschi & Stirling Identities, Asymptotics and Inequalities for the Pi (or Gamma) Function*, arXiv:2601.21906v1 (2026), https://arxiv.org/abs/2601.21906.

[2] NIST Digital Library of Mathematical Functions, Binet's formula, Eq. 5.9.10, https://dlmf.nist.gov/5.9#E10.